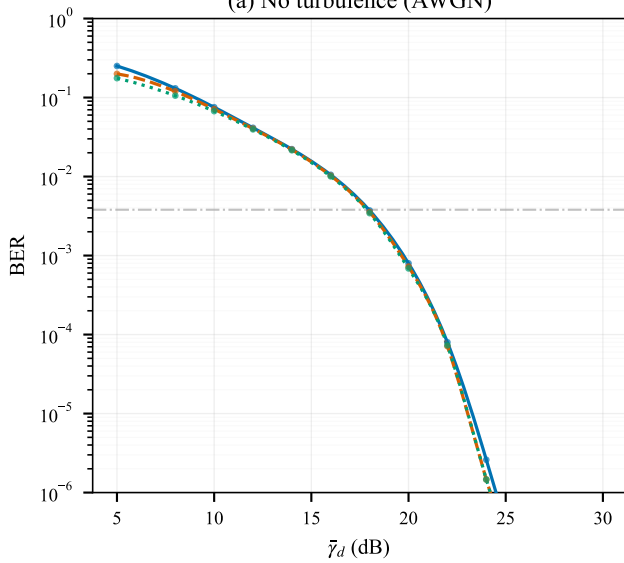
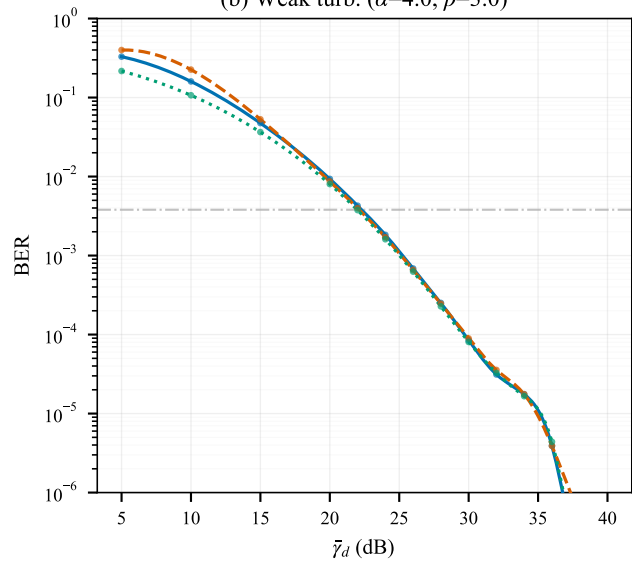
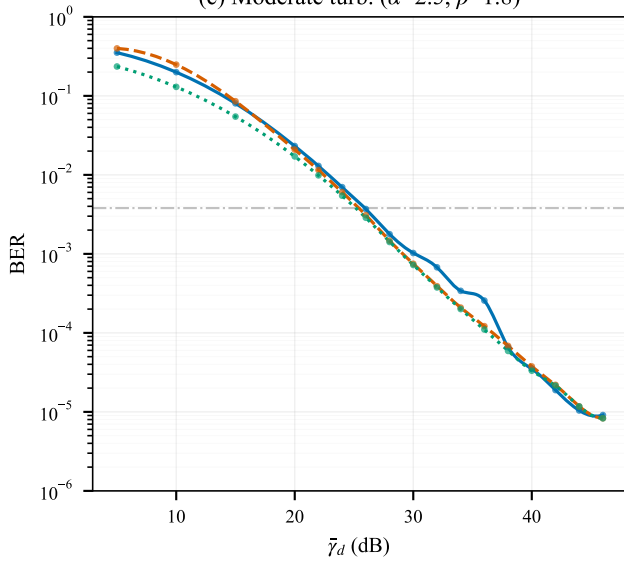
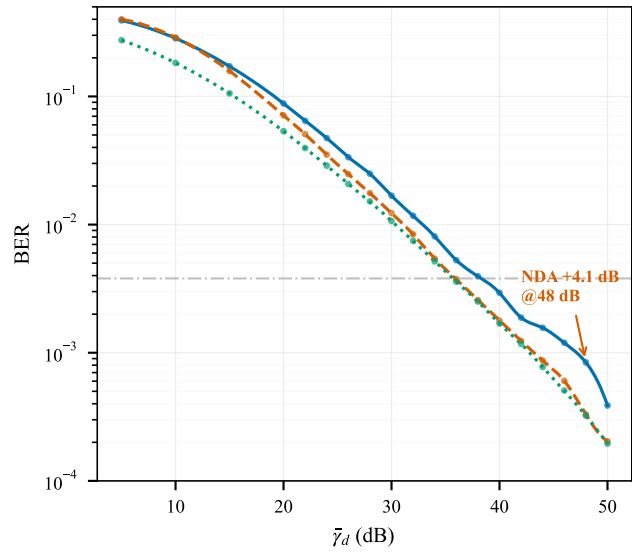
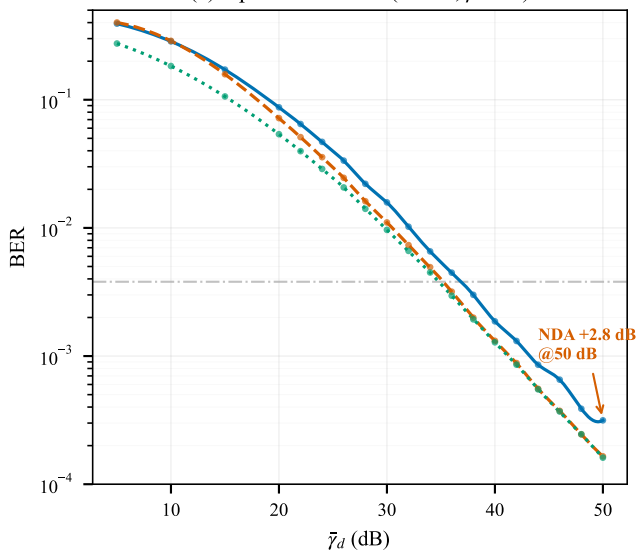
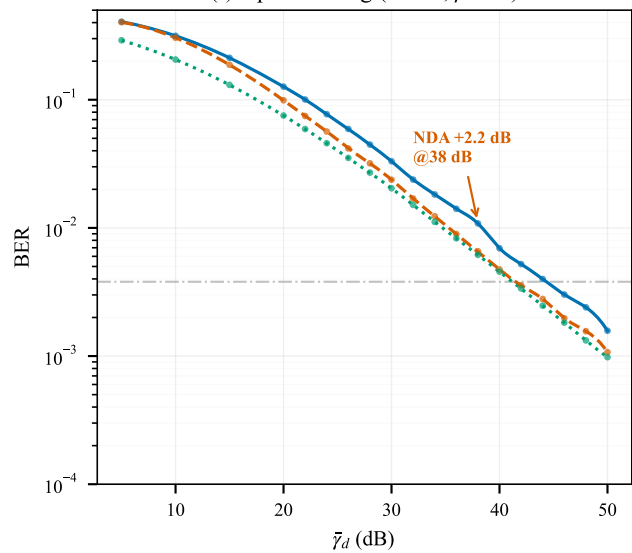


(a) No turbulence (AWGN)

(b) Weak turb. ($\alpha=4.0, \beta=3.0$)(c) Moderate turb. ($\alpha=2.5, \beta=1.8$)(d) Strong turb. ($\alpha=1.5, \beta=0.8$)(e) Uplink moderate ($\alpha=1.2, \beta=0.9$)(f) Uplink strong ($\alpha=1.0, \beta=0.7$)

— DA-ML (pilot-aided) - - - NDA-ML (blind) Oracle - - - HD-FEC (4e-03)